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Background

Long-term proton pump inhibitor (PPI) therapy carries risks and cost implications when used without a clear indication. This audit provides baseline data needed to implement strategic deprescribing efforts upon the decision of HRPB Drug and Therapeutic Committee.

Objective

To establish the **prevalence** and **prescribing patterns of long-term PPI use in adults at HRPB** and to **identify patients maintained on PPIs without a documented antiplatelet indication**.

Methodology

Study Design: Retrospective audit using pharmacy dispensing records (PhIS) and Hospital Information System.

Inclusion Criteria:
Adults dispensed with pantoprazole, lansoprazole or dexlansoprazole

Study Period:
1st January - 31st December 2024;
1st March - 31st October 2025

Collected patient MRN, prescription orders (duplicates removed), concurrent antiplatelet use, PPI agent and dose, and prescribing clinic.

Results

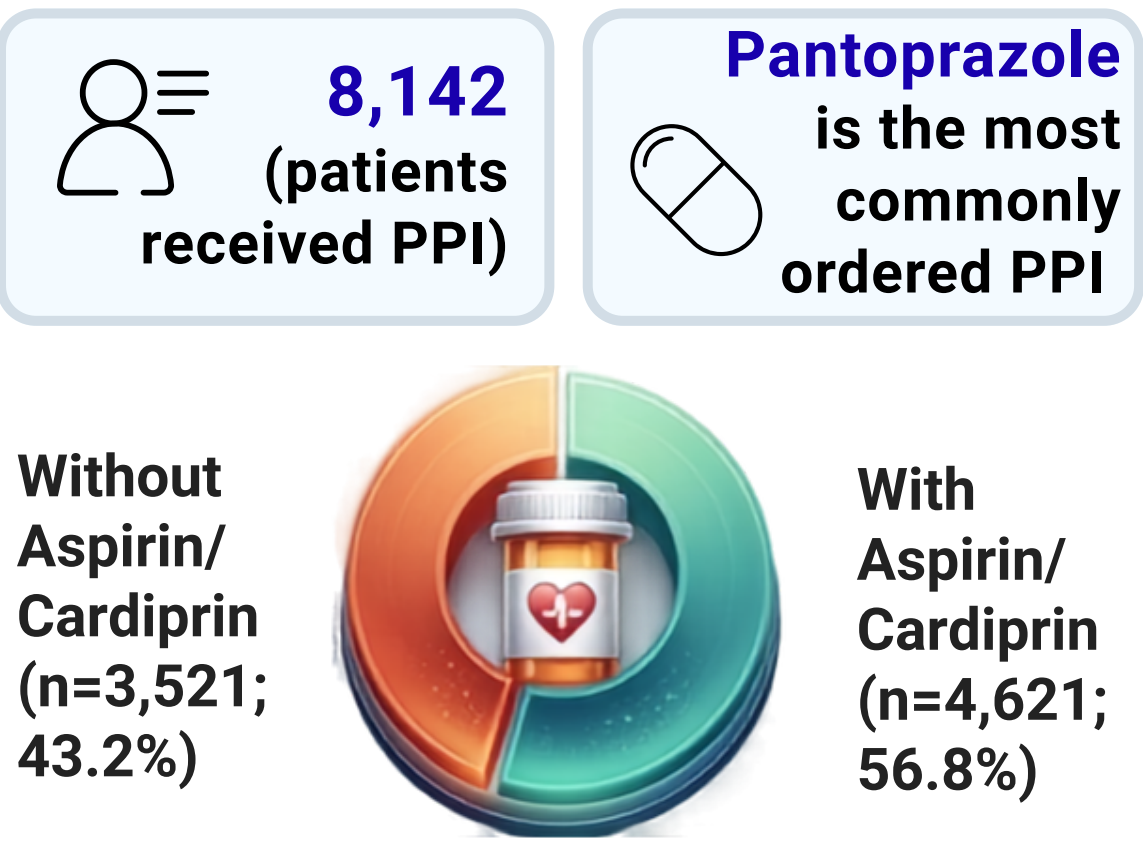


Figure 1: PPI prescribed with Antiplatelet Indication

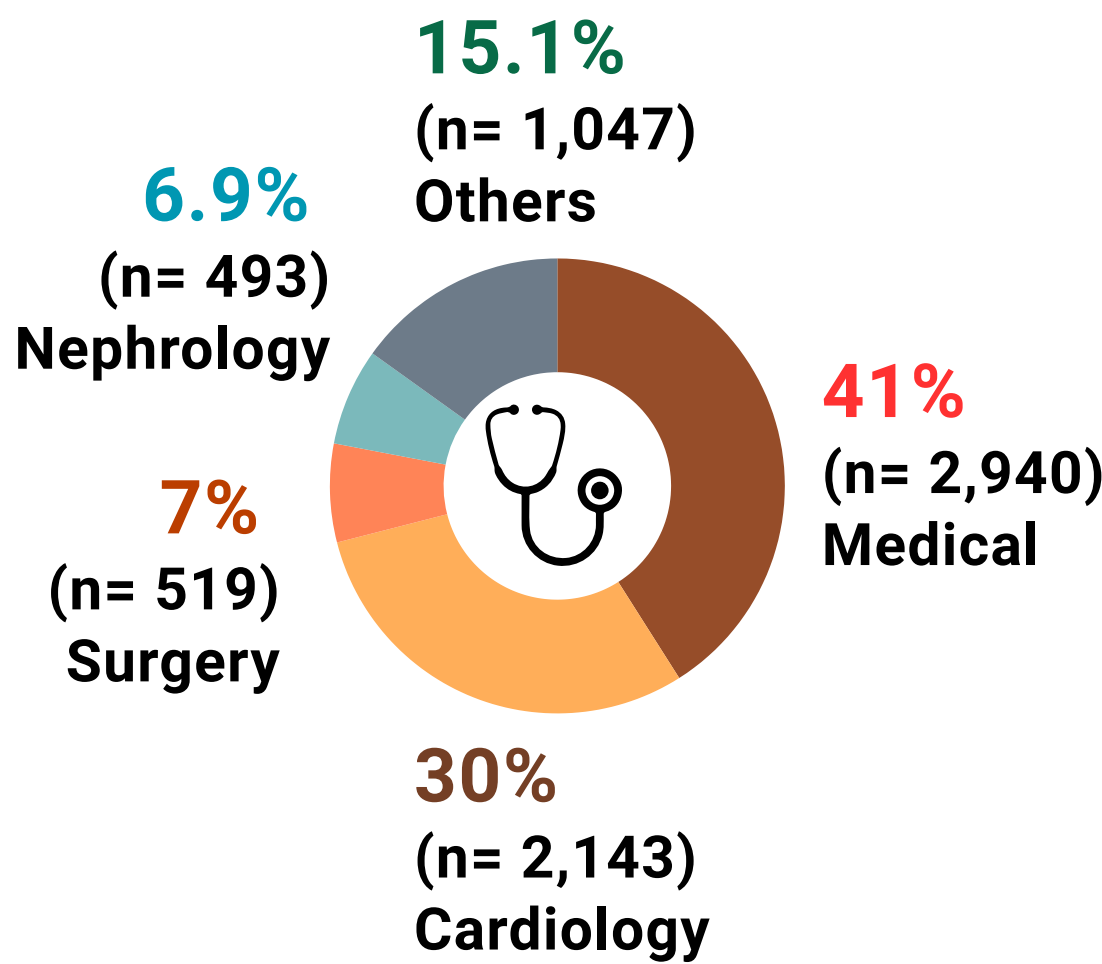


Figure 2: PPI Prescriptions by Clinics

Discussion

The use of ‘PRN’ for outpatient PPI prescriptions and **high prescribing volumes** in clinics **indicated ongoing unreviewed use**.

Continuation without review poses both **clinical safety risks** and **financial costs**.

Prolonged use of PPI disrupts natural gut defense barriers, which strongly **supports deprescribing efforts** in facility when long-term acid suppression is no longer clinically necessary.

Recommendation

Introduce **routine medication review** for patients whom on **PPI >6 months**, prioritizing **Medical** and **Cardiology Clinics**.

Implement **deprescribing protocol** with **clear stop** and **step-down guidance**, requiring **indication**, **duration of PPI**, **reduce inappropriate PRN orders** and **targeted education** to prescribers and pharmacists.

Re-audit 12 months after interventions and include **explicit appropriateness criteria**, **clinical outcomes** and **cost impact**.

Limitation

Certain valid indications (e.g. long-term NSAIDs or chronic corticosteroid therapy) might not been captured. This audit focused on antiplatelet co-prescription.